



Task No. 1 — Voltage Drop on a Distribution Transformer

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1. Specification

SN	U _{1N} / U _{2N}	Loading x	cos φ
250 kVA	22 kV / 0.4 kV	85 %	0.85 (ind. & cap.)

2. Equivalent Circuit (Shunt Branch Neglected)

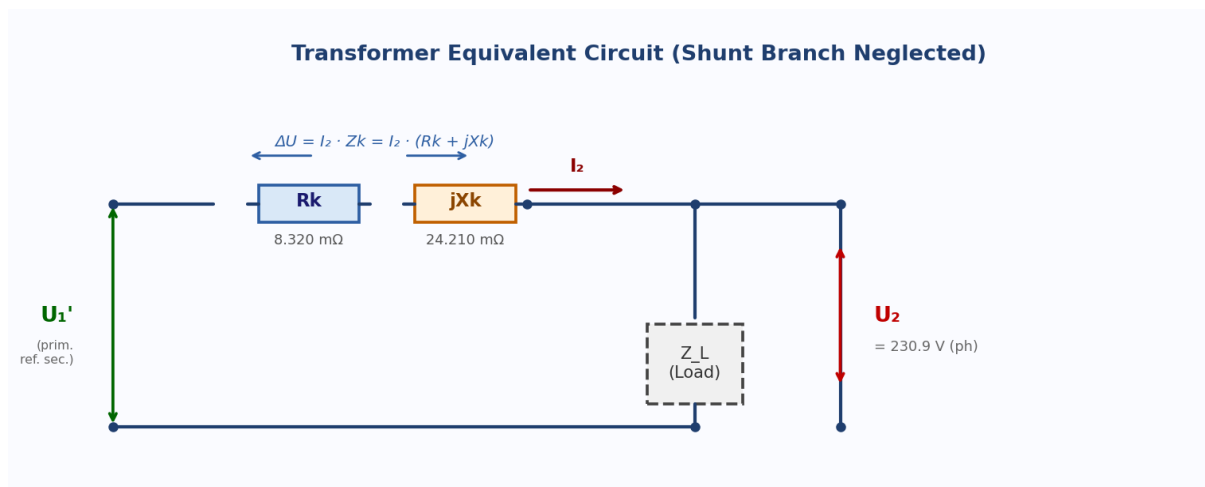


Figure 1 — Transformer equivalent circuit (per-phase, referred to secondary side). $R_k = 8.320 \text{ m}\Omega$, $X_k = 24.210 \text{ m}\Omega$.

3. Phasor Diagram

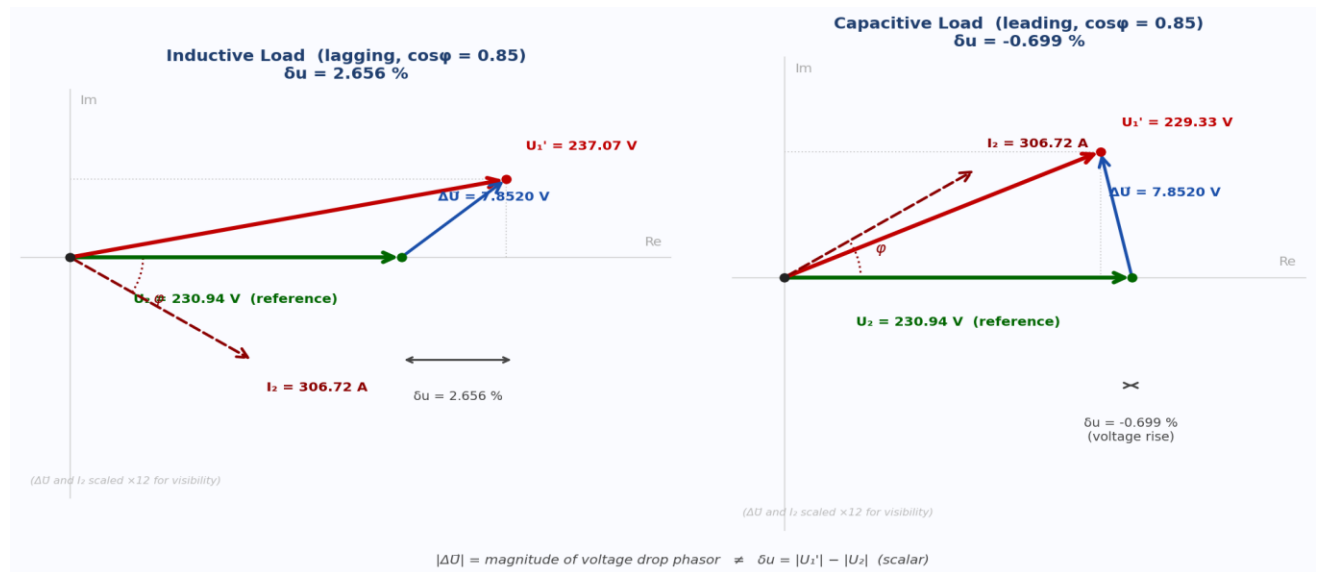


Figure 2 — Phasor diagrams for inductive (left) and capacitive (right) load. ΔU and I_2 vectors scaled $\times 12$ for visibility; all labelled values are actual.

4. Calculation

4.1 Transformer Parameters (referred to secondary)

$$\begin{aligned}
 I_{2N} &= S_N / (\sqrt{3} \cdot U_{2N}) = 250000 / (1.732 \times 400) = 360.84\text{ A} \\
 U_{2ph} &= U_{2N} / \sqrt{3} = 400 / 1.732 = 230.940\text{ V} \\
 Z_{base} &= U_{2ph}^2 / (S_N / 3) = 230.940^2 / 83333 = 0.64000\ \Omega \\
 Z_k &= u_{kr} \cdot Z_{base} = 0.04 \times 0.64000 = 0.02560\ \Omega \\
 R_k &= P_k / (3 \cdot I_{2N}^2) = 3250 / (3 \times 130208.33) = 0.008320\ \Omega \\
 X_k &= \sqrt{(Z_k^2 - R_k^2)} = 0.024210\ \Omega \\
 u_{kR} &= R_k / Z_{base} = 1.300\% \\
 u_{kX} &= X_k / Z_{base} = 3.783\% \\
 I_2 &= x \cdot I_{2N} = 0.85 \times 360.84 = 306.717\text{ A} \\
 \sin\phi &= \sqrt{1 - 0.85^2} = 0.5268
 \end{aligned}$$

4.2 Method (a) — Ohm's and Kirchhoff's Law

Reference: $U_2 \vec{=} U_{2ph} \angle 0^\circ$. Compute $U_1 \vec{=} U_2 \vec{+} I_2 \vec{\cdot} Z_k$, then $\delta u\% = (|U_1| - U_{2ph}) / U_{2ph} \times 100$

Inductive ($I_2 \vec{=} I_2 \cdot \cos\phi - j \cdot I_2 \cdot \sin\phi$):

$$\begin{aligned}
 \Delta U_{re} &= I_2 \cdot \cos\phi \cdot R_k + I_2 \cdot \sin\phi \cdot X_k = 2.1691 + 3.9117 = 6.0808\text{ V} \\
 \Delta U_{im} &= I_2 \cdot \cos\phi \cdot X_k - I_2 \cdot \sin\phi \cdot R_k = 6.3119 - 1.3443 = 4.9676\text{ V} \\
 |U_1| &= \sqrt{((230.940 + 6.0808)^2 + 4.9676^2)} = 237.0730\text{ V}
 \end{aligned}$$

$$\delta u_{a,ind} = (237.0730 - 230.940) / 230.940 \times 100 = 2.656\%$$

Capacitive ($I_2 \vec{=} I_2 \cdot \cos\phi + j \cdot I_2 \cdot \sin\phi$):

$$\begin{aligned}
 \Delta U_{re} &= I_2 \cdot \cos\phi \cdot R_k - I_2 \cdot \sin\phi \cdot X_k = 2.1691 - 3.9117 = -1.7426\text{ V} \\
 \Delta U_{im} &= I_2 \cdot \cos\phi \cdot X_k + I_2 \cdot \sin\phi \cdot R_k = 6.3119 + 1.3443 = 7.6561\text{ V} \\
 |U_1| &= \sqrt{((230.940 - 1.7426)^2 + 7.6561^2)} = 229.3253\text{ V}
 \end{aligned}$$

$$\delta u_{a,cap} = (229.3253 - 230.940) / 230.940 \times 100 = -0.699\%$$

4.3 Method (b) — Simplified Equation

$$\delta u_b\% = x \cdot (ukR \cdot \cos\phi \pm ukX \cdot \sin\phi) \cdot 100 \quad [+ \text{ inductive, } - \text{ capacitive }]$$

Inductive:

$$\delta u_{b,ind} = 0.85 \times (1.300 \times 0.85 + 3.783 \times 0.5268) = 0.85 \times 3.0977$$
$$\delta u_{b,ind} = 2.633 \%$$

Capacitive:

$$\delta u_{b,cap} = 0.85 \times (1.300 \times 0.85 - 3.783 \times 0.5268) = 0.85 \times -0.8877$$
$$\delta u_{b,cap} = -0.755 \%$$

4.4 Method (c) — Unsimplified Equation

$$\delta u_c\% = [\sqrt{((1 + x \cdot ukR \cdot \cos\phi \pm x \cdot ukX \cdot \sin\phi)^2 + (x \cdot ukX \cdot \cos\phi \mp x \cdot ukR \cdot \sin\phi)^2)} - 1] \times 100$$

Inductive:

$$P = x(ukR \cdot \cos\phi + ukX \cdot \sin\phi) = 0.026331$$
$$Q = x(ukX \cdot \cos\phi - ukR \cdot \sin\phi) = 0.021510$$
$$\delta u_{c,ind} = 2.656 \%$$

Capacitive:

$$P = x(ukR \cdot \cos\phi - ukX \cdot \sin\phi) = -0.007546$$
$$Q = x(ukX \cdot \cos\phi + ukR \cdot \sin\phi) = 0.033152$$
$$\delta u_{c,cap} = -0.699 \%$$

5. Summary Table

	Method (a) [ref]	Method (b) [simpl.]	Method (c) [exact]	Error (b) vs (a)	Error (c) vs (a)
δu Inductive	2.656 %	2.633 %	2.656 %	-0.85 %	0.000 %
δu Capacitive	-0.699 %	-0.755 %	-0.699 %	-7.92 %	0.000 %

6. Conclusion

Inductive load: all methods give $\delta u \approx 2.66 \%$. Method (b) error = 0.85 %; method (c) error = 0.000 %. Method (b) is suitable for engineering practice.

Capacitive load: $\delta u = -0.699 \%$ (negative = voltage rise). Method (b) error = 7.9 % (larger because $|\delta u|$ is small). Method (c) matches the reference to within 0.000 %.

Method (c) is the most accurate alternative. The simplified method (b) is adequate for inductive loads; for capacitive loads method (c) is recommended.